

MOYYNTY, Kazakhstan

WMO: 357930

Lat: 47.217N

Lon: 73.350E

Elev: 1919

StdP: 13.70

Time Zone: 6.00 (E06)

Period: 82-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-19.5	-14.8	-26.4	1.4	-18.2	-22.0	1.8	-13.9	31.5	26.3	22.4	20.7	1.8	0	0.581

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.9	92.9	61.7	89.8	60.9	86.9	60.1	64.6	82.7	63.3	82.3	62.1	81.4	8.5	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
60.0	82.8	68.3	57.1	74.6	68.3	54.5	67.9	67.4	30.7	82.2	29.7	82.6	28.8	81.6	73.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.7	18.7	16.3	-26.1	99.6	5.4	4.0	-30.0	102.5	-33.2	104.8	-36.3	107.0	-40.2	109.9		
(5)			-26.4	67.7	5.4	2.9	-30.3	69.8	-33.4	71.5	-36.5	73.1	-40.4	75.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	40.6	7.4	6.9	21.0	45.5	57.7	69.1	73.4	69.1	57.0	41.2	24.7	12.7		
	DBStd	25.62	10.73	10.34	10.64	9.45	7.81	7.22	5.38	5.84	7.30	9.35	10.55	10.90		
	HDD50	5883	1320	1206	898	194	27	1	0	0	22	301	759	1155		
	HDD65	9502	1785	1626	1363	587	241	44	7	28	253	739	1209	1620		
	CDD50	2471	0	0	0	58	265	573	724	592	232	27	0	0		
	CDD65	615	0	0	0	1	15	166	266	154	13	0	0	0		
	CDH74	7930	0	0	0	29	318	2122	3223	1925	299	14	0	0		
	CDH80	3022	0	0	0	4	58	822	1370	713	53	2	0	0		
	WSAvg	5.5	3.9	4.4	5.4	7.3	7.1	6.6	6.1	6.0	5.4	5.5	4.7	3.8		
	PrecAvg	12.8	1.4	1.3	1.3	1.1	1.0	1.0	1.2	0.6	0.4	0.7	1.2	1.4		
(15) Precipitation	PrecMax	27.6	3.9	2.9	3.1	2.4	3.7	3.4	3.7	1.9	1.8	2.5	2.7	3.7		
	PrecMin	6.6	0.2	0.2	0.3	0.0	0.1	0.1	0.2	0.0	0.0	0.1	0.0	0.3		
	PrecStd	5.2	0.8	0.7	0.7	0.6	0.7	0.7	0.7	0.4	0.5	0.7	0.7	0.7		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.7	32.3	54.1	77.4	85.4	95.2	97.9	95.7	86.3	74.7	52.8	37.2		
		MCWB	30.7	30.7	46.1	54.0	57.9	62.1	63.4	62.1	58.2	53.1	43.8	33.9		
	2%	DB	27.8	29.5	45.8	71.8	81.3	90.9	93.5	91.7	81.0	67.4	45.8	32.6		
		MCWB	26.5	28.3	40.8	51.6	56.5	60.8	62.5	60.5	55.4	49.4	40.5	30.7		
	5%	DB	25.1	26.6	39.4	67.4	77.9	88.0	90.3	87.5	77.7	61.8	41.5	30.2		
		MCWB	24.0	25.5	36.3	50.0	55.2	60.2	61.5	59.3	54.4	46.5	38.0	28.7		
	10%	DB	22.3	23.3	34.8	62.6	74.2	84.9	87.5	84.2	73.8	56.7	37.0	27.8		
		MCWB	21.3	21.8	32.6	47.6	53.7	59.1	61.1	58.4	52.8	44.7	33.7	26.5		
	0.4%	WB	30.6	31.4	47.6	56.3	61.4	65.9	67.3	64.7	59.3	54.4	45.1	34.0		
		MCDB	31.4	31.8	52.1	72.2	76.7	85.9	80.1	88.2	81.2	68.8	49.9	35.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	27.0	28.4	41.4	53.6	59.1	63.9	65.3	62.4	56.8	51.0	40.9	31.2		
		MCDB	27.7	29.5	44.6	67.5	74.7	81.9	81.8	84.3	78.1	63.7	44.0	32.4		
	5%	WB	24.2	25.4	36.8	51.3	57.1	62.6	63.9	61.0	55.1	48.1	38.0	29.0		
		MCDB	24.9	26.6	39.6	63.7	72.9	81.1	82.3	81.3	75.0	59.9	41.7	30.3		
	10%	WB	21.3	21.9	32.9	48.8	55.5	60.9	62.9	59.9	53.4	45.6	34.1	26.6		
		MCDB	22.2	23.2	34.8	61.2	70.8	79.7	82.0	79.7	71.9	55.6	36.3	27.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.3	16.6	22.1	24.1	24.5	23.9	25.0	25.7	20.0	15.4	15.6		
		MCDBR	15.6	14.2	18.1	28.9	29.9	28.3	27.7	29.8	30.7	29.0	18.8	15.3		
		MCWBR	15.1	13.2	13.1	13.6	12.3	10.0	9.8	11.1	14.3	15.8	13.6	13.6		
	5% WB	MCDBR	15.3	14.2	16.8	25.4	24.9	25.2	23.5	25.8	28.0	25.7	17.3	13.6		
		MCWBR	15.0	13.4	12.6	13.3	11.5	10.0	8.6	10.3	13.7	15.1	13.2	12.6		
(40) Clear-Sky Solar Irradiance	taub		0.299	0.329	0.379	0.419	0.427	0.420	0.432	0.387	0.343	0.343	0.329	0.301		
	taud		2.075	2.008	2.050	2.080	2.113	2.197	2.176	2.292	2.375	2.295	2.207	2.132		
	Ebn at Noon		238	256	262	263	266	268	263	271	274	252	227	221		
	Edh at Noon		30	39	45	48	48	44	45	38	32	30	26	25		
(44) All-Sky Solar Radiation	RadAvg		544	883	1277	1695	2114	2225	2102	1969	1538	925	506	390		
	RadStd		69	105	103	136	125	125	142	114	89	83	53	55		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (3 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page